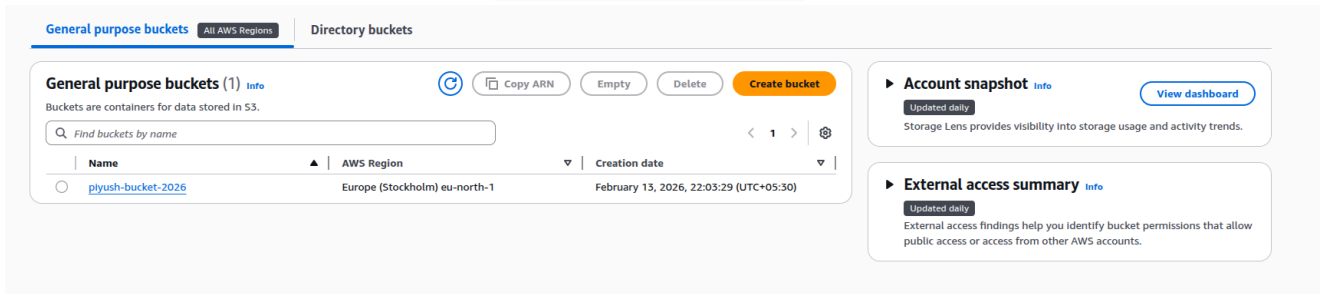


1. Create an S3 bucket with a unique name and select an appropriate region.

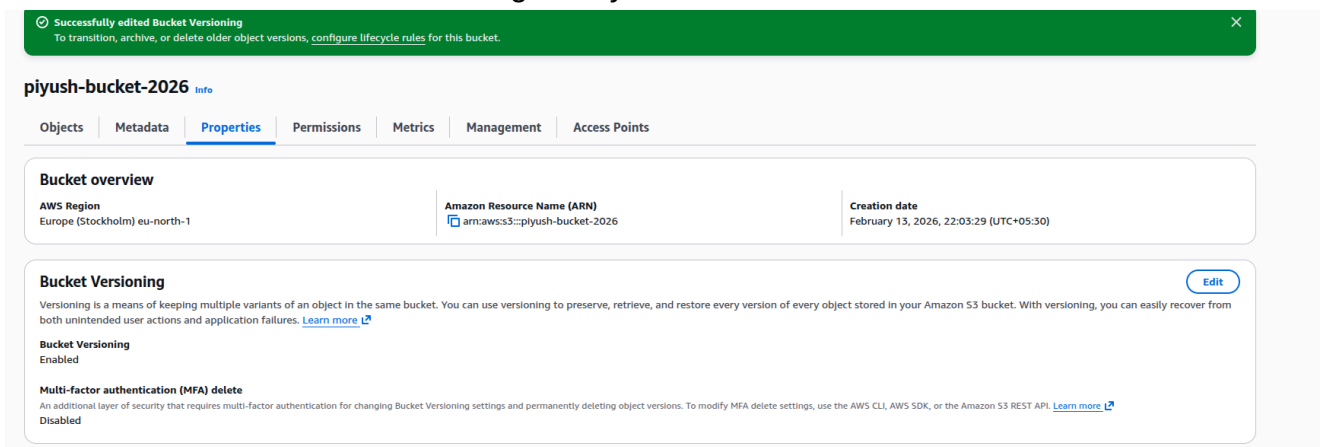
Ans:

I created an S3 bucket with name `piyush-bucket-2026` in europe stockholm region.

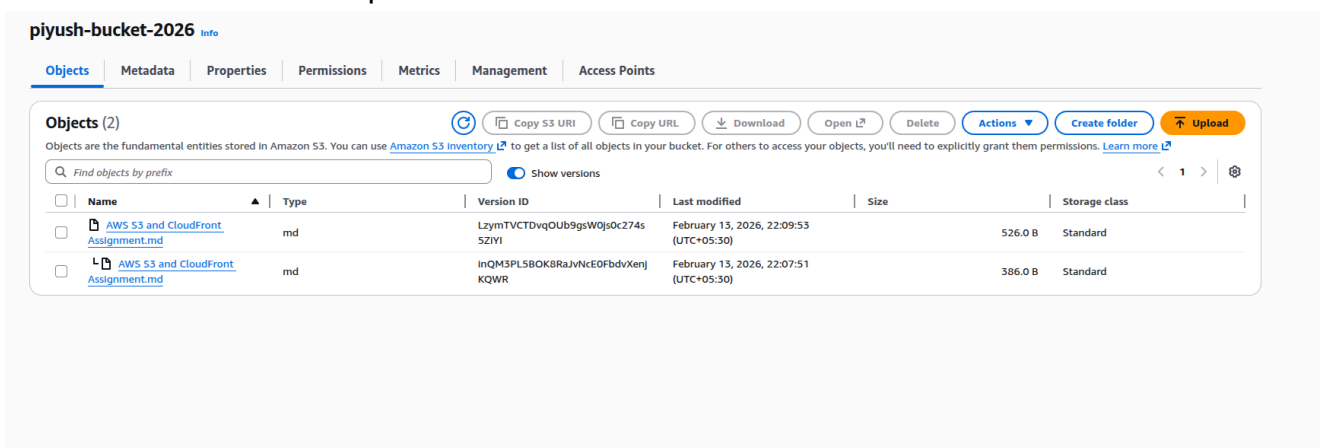


2. Enable bucket versioning and upload multiple versions of the same file.

Ans: I enabled the bucket versioning for my bucket.



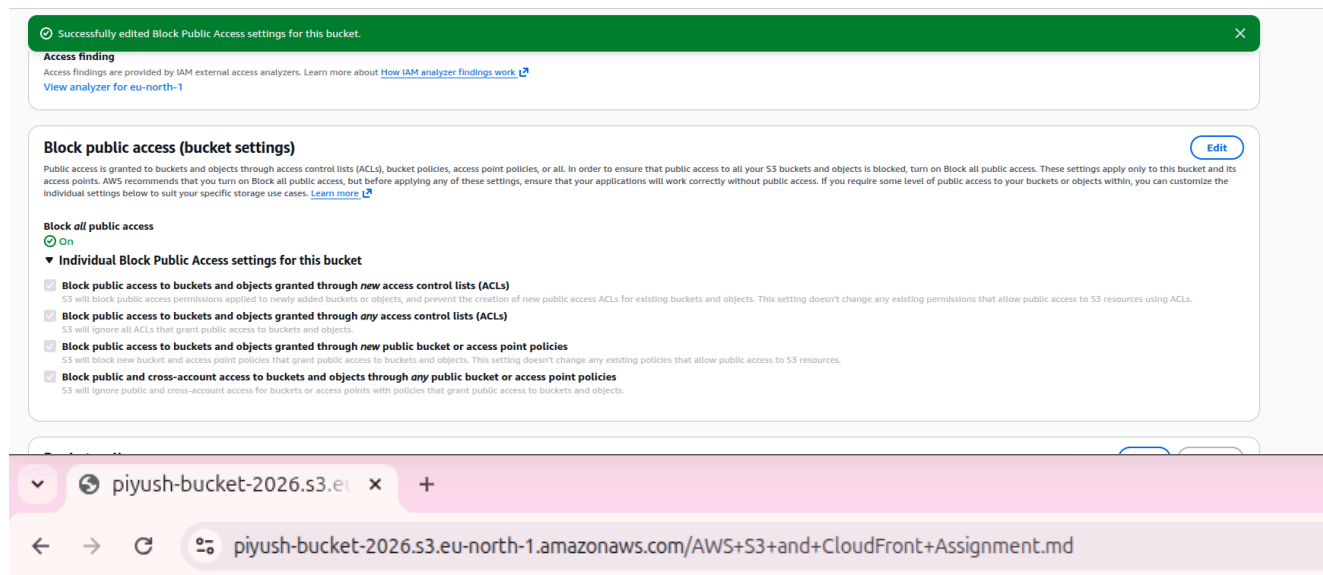
Now I uploaded multiple versions of the same file and observed that file had different versions created with unique version id and details.



3. Configure bucket-level block public access settings and verify restrictions.

Ans:

I configured the bucket-level block public access settings to block all the access , Now verifying the restrictions as well by making the object public but it will give access denied.



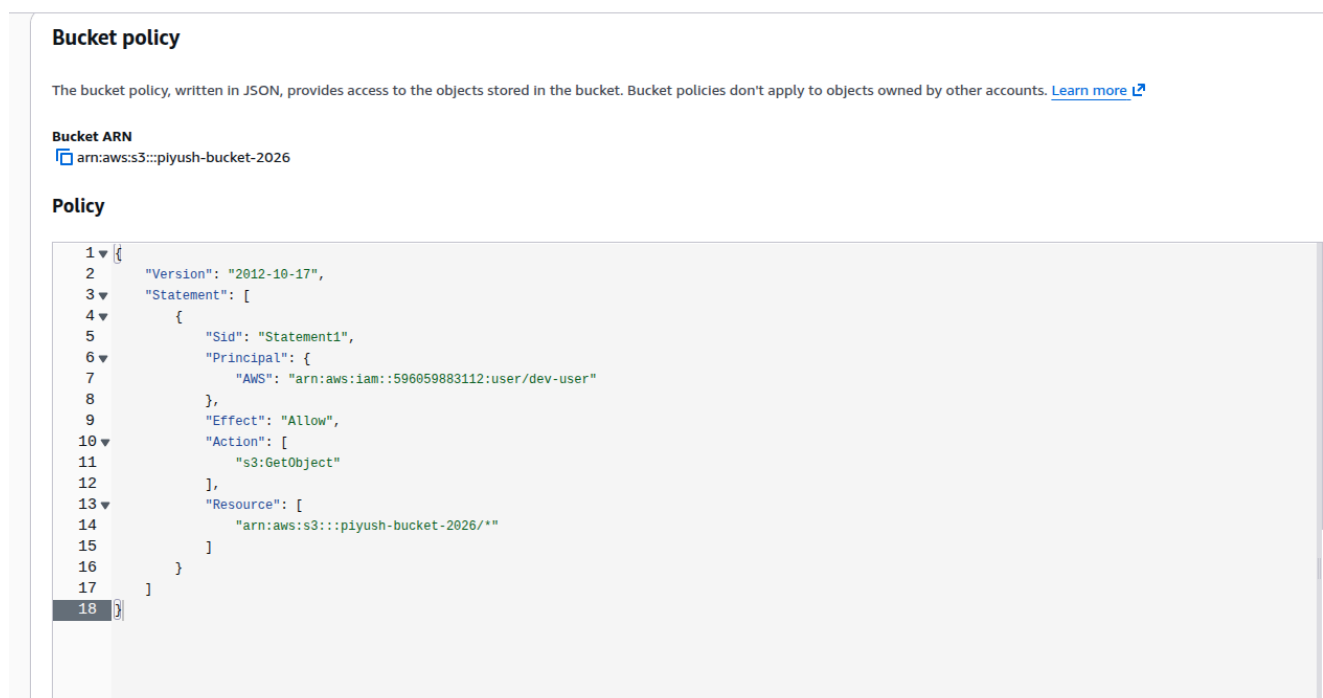
This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>4YY933KTNWMF2SYG</RequestId>
  <HostId>J0INBDehnE/lbte9moxKtTZuuKvqi6KhPD4iAaJq0dYkqz3WA0y41TvFLPi5wS0oGp94bsQYInY=</HostId>
</Error>
```

4. Create a bucket policy that allows read access only from a specific IAM user or role.

Ans:

This is the policy I created to allow read access only to a specific user or role.



Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

arn:aws:s3:::piyush-bucket-2026

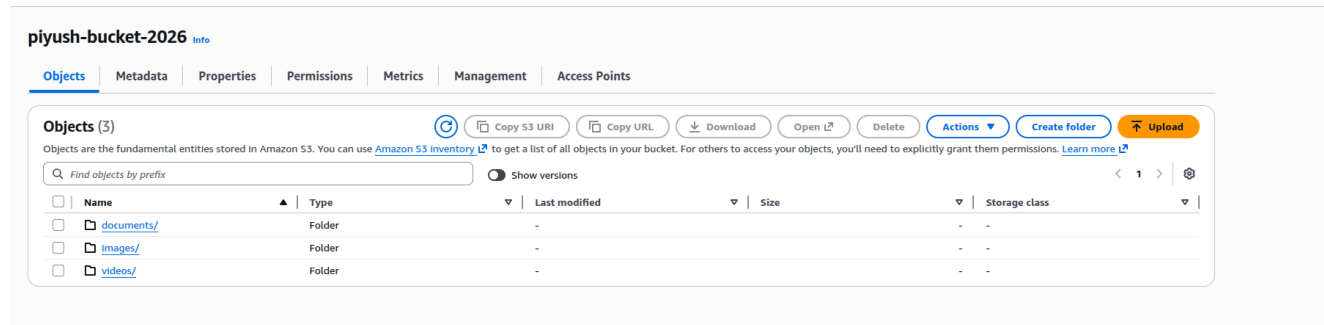
Policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Statement1",
      "Principal": {
        "AWS": "arn:aws:iam::596059883112:user/dev-user"
      },
      "Effect": "Allow",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::piyush-bucket-2026/*"
      ]
    }
  ]
}
```

5. Upload different file types (images, videos, documents) and organize them using folders/prefixes.

Ans:

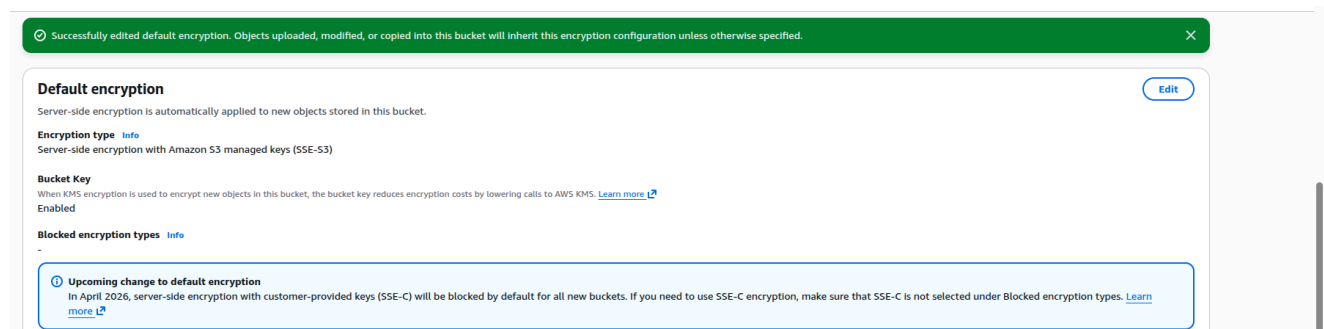
I created first three folders and then uploaded different files in different folders.



6. Enable default encryption on the bucket (SSE-S3 or SSE-KMS).

Ans :

I added default encryption on the bucket of type SSE-S3 .



7. Configure lifecycle rules to:

- Transition objects to Glacier after 30 days
- Expire objects after 90 days
- Enable server access logging and store logs in a separate bucket.

Ans:

I created a lifecycle rule titled `my-lifecycle-rule` and added the asked rules in the lifecycle rules .

my-lifecycle-rule

Info

EditDeleteActions

Lifecycle rule configuration

Lifecycle rule name

my-lifecycle-rule

Status

Enabled

Scope

Entire bucket

Prefix

-

Object tags

-

Minimum object size

-

When no minimum object size is specified, the minimum object size for transitions is determined by the lifecycle configuration. [Learn more](#)

Maximum object size

-

Review transition and expiration actions

Current version actions

Day 0

Objects uploaded

↓

Day 30

Objects move to Glacier Instant Retrieval

↓

Day 90

Objects expire

Noncurrent versions actions

Day 0

No actions defined.

I enabled server access logging and use different bucket to store logs in a separate bucket.

Successfully edited server access logging.

Enable objects stored in the Intelligent-Tiering storage class to tier-down to the Archive Access tier or the Deep Archive Access tier which are optimized for objects that will be rarely accessed for long periods of time. [Learn more](#)

Find Intelligent-Tiering Archive configurations

Name	Status	Scope	Days until transition to Archive Access...	Days until transition to Deep Archive ...
No archive configurations				
No configurations to display.				

Create configuration

Server access logging

Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)

Server access logging

Enabled

Destination bucket

s3://plyush-bucket-2026

Log object key format

[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

Edit

AWS CloudTrail data events

You can view and configure CloudTrail data events for Amazon S3 bucket object-level operations in the AWS CloudTrail console. [AWS CloudTrail](#)

Event notifications (0)

Send a notification when specific events occur in your bucket. [Learn more](#)

EditDeleteCreate event notification

Name	Event types	Filters	Destination type	Destination
------	-------------	---------	------------------	-------------

Use this page to e requests to access Server access logs detailed records fr are made to a buc provide insight in are receiving. Ser useful for many a including underst access, and your #

By default, Amaz collect server acc Enable to turn on the Destination w receive the log re This bucket must Region as the sou

When you enable logging for the bu S3 console autom your bucket polic to the S3 log deli

Was this cont

Yes

Learn more

Logging bucket re

8. Generate a pre-signed URL and access a private object using it.

Ans:

I generated a pre-signed URL to securely share a private object stored in my S3 bucket. Since the bucket had Block Public Access enabled, objects were not publicly accessible. Using the pre-signed URL feature, I granted temporary read access to a specific object. I configured an expiration time for the URL and successfully accessed the private file through the generated link in my browser.

A presigned URL for "image.webp" has been created and copied to your clipboard.

Copy presigned URL

images/

Copy S3 URI

Objects

Properties

Objects (1/1)

Copy S3 URICopy URLDownloadOpenDeleteActionsCreate folderUpload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

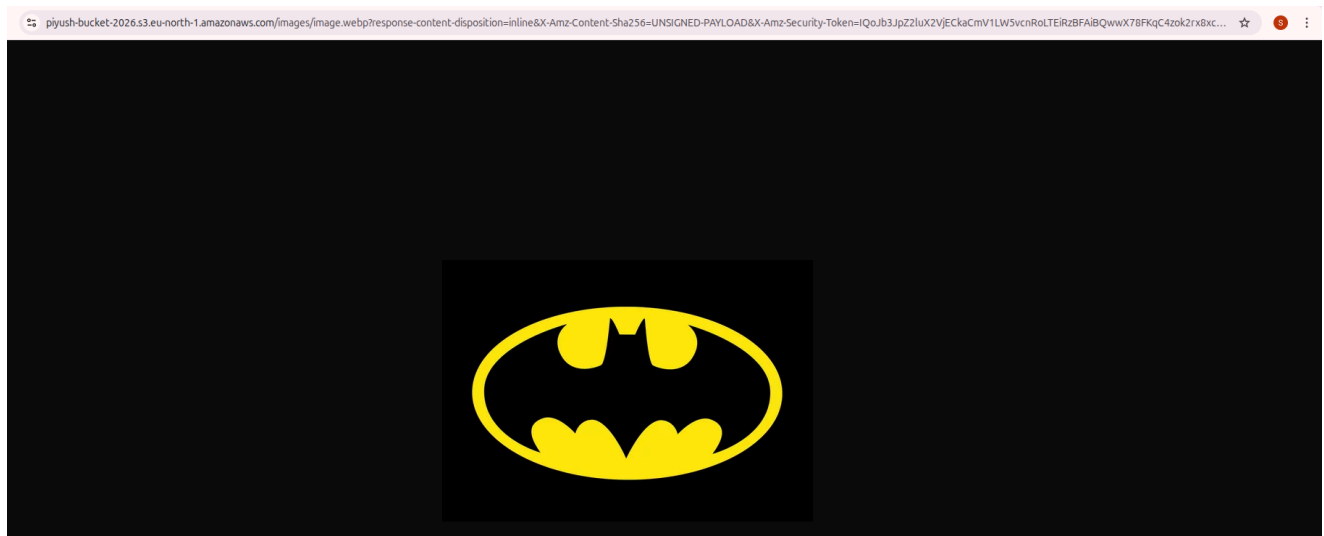
Find objects by prefix

Show versions

Name	Type	Last modified	Size	Storage class
image.webp	webp	February 13, 2026, 22:31:15 (UTC+05:30)	9.6 KB	Standard

I was able to view my image which i uploaded in the images folder of my bucket using the

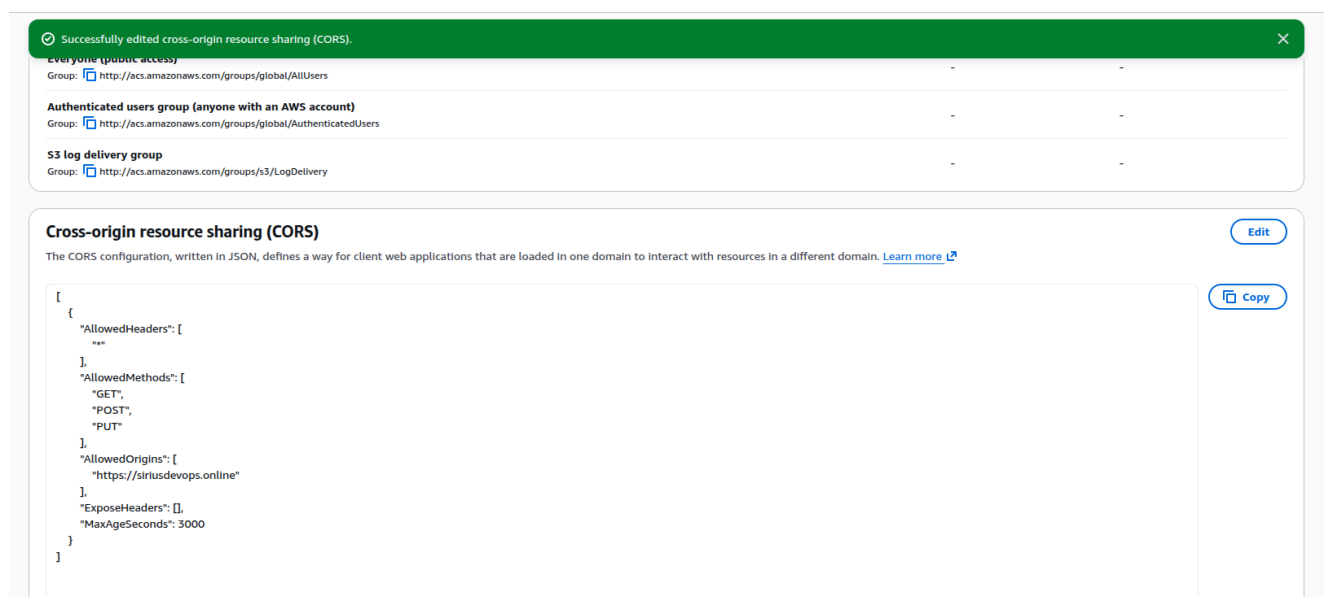
pre-signed-url.



9. Configure CORS to allow access from a specific domain.

Ans:

I configured Cross-Origin Resource Sharing (CORS) on my S3 bucket to allow web applications from a specific domain to access bucket resources. This is useful when hosting frontend applications that need to fetch images, videos, or documents from S3. I added a CORS policy specifying the allowed origin, HTTP methods, and headers. After saving the configuration, cross-origin requests from the specified domain were permitted.



10. Enable static website hosting and access the website via the S3 endpoint.

Ans:

I enabled static website hosting on my S3 bucket to host a simple website. I uploaded HTML files (index and error pages) and configured the bucket for website hosting. After enabling the feature, AWS generated a website endpoint URL. I accessed the endpoint in my browser and verified that my static website was successfully hosted.

Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://piyush-bucket-2026

Succeeded

2 files, 254.0 B (100.00%)

Failed

0 files, 0 B (0%)

Files and folders
Configuration

Files and folders (2 total, 254.0 B)

Find by name

Name	Folder	Type	Size	Status	Error
error.html	-	text/html	93.0 B	Succeeded	-
index.html	-	text/html	161.0 B	Succeeded	-

Successfully edited static website hosting.

Object Lock

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

Object Lock
Disabled

Requester pays

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
Disabled

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using [AWS Amplify Hosting](#) for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

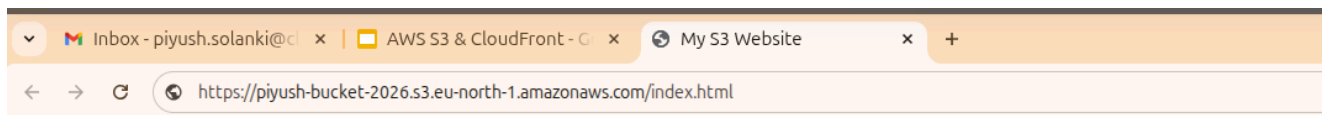
S3 static website hosting
Enabled

Hosting type
Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://piyush-bucket-2026.s3-website-eu-north-1.amazonaws.com>



Welcome to My Static Website

Hosted on Amazon S3

11. Simulate accidental deletion and restore a previous version of an object.

Ans:

I simulated accidental deletion of an object in my version-enabled S3 bucket. Since versioning was enabled, Amazon S3 did not permanently delete the object but instead created a delete marker. I accessed the object's version history, removed the delete marker,

and restored the previous version successfully. This demonstrated how S3 versioning protects data from accidental loss.

I deleted index.html .

The screenshot shows the AWS S3 console interface. At the top, a green banner indicates "Successfully deleted objects" with a link to "View details below." Below this, the "Delete objects: status" section shows a summary of the deletion process. The "Summary" section indicates that 1 object (161.0 B) was successfully deleted, while 0 objects failed to delete. The "Failed to delete" section is currently empty, showing 0 objects. Below the summary, the "Configuration" tab is selected, showing the bucket's configuration. The "Objects" tab is also visible, showing a list of objects in the bucket. The list includes "error.html" and "index.html", both of which are 161.0 B in size and were last modified on February 13, 2026, at 22:51:34 (UTC+05:30). The "index.html" object is highlighted, indicating it was the object that was deleted.

Delete objects: status

After you navigate away from this page, the following information is no longer available.

Summary

Source: s3://piyush-bucket-2026

Successfully deleted: 1 object, 161.0 B

Failed to delete: 0 objects

Failed to delete | Configuration

Failed to delete (0)

Find objects by name

Name	Folder	Type	Last modified	Size	Error
No objects failed to delete.					

piyush-bucket-2026

Objects | Metadata | Properties | Permissions | Metrics | Management | Access Points

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

Name	Type	Last modified	Size	Storage class
error.html	html	February 13, 2026, 22:51:33 (UTC+05:30)	93.0 B	Standard
index.html	html	February 13, 2026, 22:51:34 (UTC+05:30)	161.0 B	Standard

12. Upload a large file using multipart upload.

Ans:

I uploaded a large file to my S3 bucket using the multipart upload feature. Multipart upload allows large objects to be uploaded in smaller parts, improving upload speed and reliability. If any part fails, only that part needs to be re-uploaded. I uploaded a large file (greater than 100 MB) and verified that the multipart upload completed successfully.

The screenshot shows the Amazon S3 console interface during a multipart upload. The "Uploading" section shows a progress bar at 19%. The "Server-side encryption" section indicates that no encryption key is specified, and the bucket's default settings are used. The "Checksums" section shows that checksums are used for data integrity verification, and the "Checksum function" is set to CRC64NVME. The "Tags - optional" section shows that no tags are associated with the resource.

Uploading

19%

Total remaining: 95 files; 22.5 MB (80.51%)

Estimated time remaining: 2 minutes

Transfer rate: 160.5 KB/s

Server-side encryption

No encryption key specified

The bucket settings for default encryption are used to encrypt objects when storing them in Amazon S3.

Checksums

Checksums are used for data integrity verification of new objects. [Learn more](#)

Checksum function

CRC64NVME

Tags - optional (0)

You can use object tags to analyze, manage, and specify permissions for objects. [Learn more](#)

Key	Value
No tags associated with this resource.	

13. Estimate monthly storage cost using AWS Pricing Calculator.

Ans:

I used the AWS Pricing Calculator to estimate the monthly storage cost of my S3 bucket. I configured storage parameters such as storage class, amount of data stored, number of requests, and data transfer. Based on my estimated usage, the calculator generated an approximate monthly cost. This helped me understand the pricing model of Amazon S3 and plan cost optimization strategies.

Pricing Calculator > Saved estimates > MyS3Cost

MyS3Cost Export Settings

Workload estimate allows you to estimate the cost of specific workloads, applications, resources, and architectural changes.

Usage (5) Edit Delete Add

Filter by: Account / Service / Region / Cost / State / Group Group by: Region

☐	Service > Usage lines	Group	Location	Account	Usage amount	State	Historical cost	Estimated cost
☐	▼ EU (Stockholm) eu-north-1							\$0.52 USD
☐	EUN1-Select-Returned-Bytes	MyCostGroup	EU (Stockholm) eu-n...	596059883112	20 GB	🟢 New	-	\$0.01 USD
☐	EUN1-Requests-Tier1	MyCostGroup	EU (Stockholm) eu-n...	596059883112	1500 Requests	🟢 New	-	\$0.01 USD
☐	EUN1-TimedStorage-ByteHrs	MyCostGroup	EU (Stockholm) eu-n...	596059883112	20 GB-Mo	🟢 New	-	\$0.46 USD
☐	EUN1-Select-Scanned-Bytes	MyCostGroup	EU (Stockholm) eu-n...	596059883112	20 GB	🟢 New	-	\$0.04 USD
☐	EUN1-Requests-Tier2	MyCostGroup	EU (Stockholm) eu-n...	596059883112	1500 Requests	🟢 New	-	\$0.00 USD

AWS CloudFront Assignment

1. Create a CloudFront distribution with an S3 bucket as the origin and host a static website over HTTPS.

Ans:

I created an Amazon S3 bucket to store my static website files such as HTML, CSS, JavaScript, and images. While creating the bucket, I kept Block Public Access enabled to ensure the bucket remained private. After uploading my static website files, I created a CloudFront distribution and selected my S3 bucket as the origin. Instead of using the static website endpoint, I used the S3 REST endpoint so that I could restrict direct public access.

Europe (Stockholm)
SiriusDEV (5960-5988-3112)
root

Amazon S3 > Buckets > piyush-bucket-2026-1

piyush-bucket-2026-1 Info

Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points

Objects (2)

Refresh
Copy S3 URI
Copy URL
Download
Open
Delete
Actions

Create folder
Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

<
1
>
Settings

☐	Name	Type	Last modified
☐	error.html	html	February 14, 2026, 20:50:07 (UTC+)
☐	index.html	html	February 14, 2026, 20:50:08 (UTC+)

MyDistribution Standard

View metrics

Details

Distribution domain name d2v5y396zr42wz.cloudfront.net	Billing Pay-as-you-go Switch to a plan	ARN arn:aws:cloudfront::596059883112:distribution/E1P77O3NBUSMMU	Last modified Deploying
--	--	--	--

General
Security
Origins
Behaviors
Error pages
Invalidations
Logging
Tags

Enable security protections to view the full security dashboard for this distribution

Gain visibility into top security trends, allowed and blocked traffic, as well as visibility and controls for bots. You'll later have the option to enable security logs to investigate traffic patterns—visually search, filter, and inspect individual log lines without writing queries. [Learn more](#)

Manage security protections

Security - Web Application Firewall (WAF) Info

Keep your application secure from the most common web threats and security vulnerabilities using AWS WAF. Blocked requests are stopped before they reach your web servers.

Core protections
Not enabled

Advanced DDoS protection
Not enabled

Manage protections

2. Configure the S3 bucket to block all public access and allow access only through CloudFront (using OAI/OAC).

Ans:

To secure my static website, I configured the S3 bucket so that it blocks all public access and allows content delivery only through CloudFront.

First, I ensured that the S3 bucket had Block Public Access enabled at the bucket level. This prevented any direct public access to objects via S3 URLs.

Next, while creating the CloudFront distribution, I configured Origin Access Control (OAC) and associated it with my S3 origin. OAC signs requests made by CloudFront before accessing the S3 bucket.

Amazon S3

▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

▼ Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

▼ Storage management and insights

Storage Lens

Batch Operations

Account and organization settings

► AWS Marketplace for S3

Bucket policy

[Edit](#)[Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. [Learn more about using Amazon S3 Block Public Access](#)

```
{
  "Version": "2008-10-17",
  "Id": "PolicyForCloudFrontPrivateContent",
  "Statement": [
    {
      "Sid": "AllowCloudFrontServicePrincipal",
      "Effect": "Allow",
      "Principal": {
        "Service": "cloudfront.amazonaws.com"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::piyush-bucket-2026-1/*",
      "Condition": {
        "ArnLike": {
          "AWS:SourceArn":
            "arn:aws:cloudfront::596059883112:distribution/E1P77O3NBUSM
            MU"
        }
      }
    }
  ]
}
```

[Copy](#)

CloudFront

Distributions

Policies

Functions

Static IPs

VPC origins

▼ SaaS

Multi-tenant distributions

Distribution tenants

▼ Telemetry

Monitoring

Alarms

Logs

▼ Reports & analytics

Cache statistics

Popular objects

Top referrers

Usage

Viewers

▼ Security

Origin access

Trust stores

Field-level encryption

--

Origin domain

Choose an AWS origin, or enter your origin's domain name. [Learn more](#)

piyush-bucket-2026-1.s3.eu-north-1.amazonaws.com

Enter a valid DNS domain name, such as an S3 bucket, HTTP server, or VPC origin ID.

Origin path - optional

Enter a URL path to append to the origin domain name for origin requests.

Enter the origin path

Name

Enter a name for this origin.

piyush-bucket-2026-1.s3.eu-north-1.amazonaws.com-mlmgs8qh4y1

Origin access

[Info](#)

☐ Public

Bucket must allow public access.

☒ Origin access control settings (recommended)

Bucket can restrict access to only CloudFront.

☐ Legacy access identities

Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access control

Select an existing origin access control (recommended) or create a new control.

oac-piyush-bucket-2026-1.s3.eu-north-1.amazonaws.com-mlmgsxzg59n

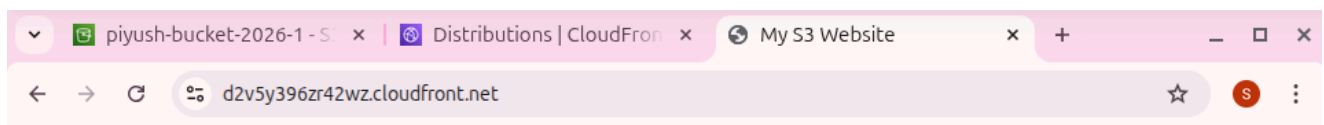
[Create new OAC](#)

✓ Policy statement copied

ⓘ You must allow access to CloudFront using this policy statement. [Learn more about giving CloudFront permission to access the S3 bucket](#)

[Copy policy](#)

Go to S3 bucket permissions



Welcome to My Static Website

Hosted on Amazon S3

3. Configure cache behaviors so that:

/static/ → 24-hour caching

/api/ → 60-second caching

/login/ → caching disabled*

Ans:

I configured multiple cache behaviors in my CloudFront distribution to control caching based on URL path patterns. I created separate behaviors for `/static/*`, `/api/*`, and `/login/*`.

For `/static/*`, I set the cache TTL to 24 hours so that static assets like CSS, JS, and images are cached longer.

For `/api/*`, I set the TTL to 60 seconds to allow short-term caching of dynamic responses.

For `/login/*`, I disabled caching completely to ensure sensitive authentication responses are always fetched from the origin. After configuring the behaviors, I verified caching using response headers such as `X-Cache` and `Age`.

CloudFront > Distributions > E1P77O3NBUSMMU

Successfully created new cache behavior /login/*.

MyDistribution

Standard

View metrics

Details

Distribution domain name

d2v5y396zr42wz.cloudfront.net

Billing

Pay-as-you-go

Switch to a plan

ARN

arn:aws:cloudfront::596059883112:distribution:E1P77O3NBUSMMU

Last modified

Deploying

General

Security

Origins

Behaviors

Error pages

Invalidations

Logging

Tags

Behaviors (4)

Save

Move up

Move down

Edit

Delete

Create behavior

Filter behaviors by property or value

	Preced...	Path patt...	Origin or ...	Viewer pr...	Cache policy name	Origin req...	R
☐	0	/static/*	piyush-bu...	Redirect H...	static-policy	-	-
☐	1	/api/*	piyush-bu...	Redirect H...	api-policy	-	-
☐	2	/login/*	piyush-bu...	Redirect H...	Managed-CachingDisabled	-	-
☐	3	Default (*)	piyush-bu...	Redirect H...	Managed-CachingOptimized	-	-

Welcome to My Static Website

Hosted on Amazon S3

Elements

Console

Sources

Network >>

1

Preserve log

Disable cache

No throttling

Filter

All

Fetch/XHR

Doc

CSS

JS

Font

Img

Media

Manifest

Socket

Wasm

Other

Name	X	Headers	Preview	Response	Initiator	Timing
d...		General				
fa...		Request URL		https://d2v5y396zr42wz.cloudfront.net/		
		Request Method		GET		
		Status Code		304 Not Modified		
		Remote Address		13.224.161.26:443		
		Referrer Policy		strict-origin-when-cross-origin		
		Response Headers				
		Age		360		
		Date		Sat, 14 Feb 2026 16:12:56 GMT		
		Etag		"38be826f88f3f83b8fb118b9b32d64cc"		
		Server		AmazonS3		
		Via		1.1 e2b5386e09b1698ae0c3be879b94940a.cloudfront.net (CloudFront)		
		X-Amz-Cf-Id		bUvjBeUSZEiUw2OQoGt_ru82RZr7DXpX3px4g_xDOO42Ju1IAAYS0A==		
		X-Amz-Cf-Pop		DEL51-P5		
		X-Amz-Server-Side-Encoding		AES256		
		X-Cache		Hit from cloudfront		
		Request Headers				
		:authority		d2v5y396zr42wz.cloudfront.net		

4. Upload static files to S3 and verify whether responses are served from cache or origin using response headers.

Ans:

I uploaded static website files such as HTML, CSS, JavaScript, and images to my private S3 bucket.

After deployment, I accessed the files through the CloudFront domain URL. To verify whether content was served from cache or origin, I inspected response headers in the browser developer tools. I observed headers like `X-Cache` and `Age`.

Initially, requests showed Miss from CloudFront, indicating content was fetched from the origin.

On subsequent requests, the header changed to **Hit from CloudFront**, confirming the content was served from cache.

The screenshot shows a web browser displaying a static website with the heading "Welcome to My Static Website" and the text "Hosted on Amazon S3". To the right, the browser's developer tools are open to the Network tab, showing a list of requests. The selected request is a GET request to a CloudFront URL. The response headers are expanded, showing the following information:

General	
Request URL	https://d2v5y396zr42wz.cloudfront.net/
Request Method	GET
Status Code	304 Not Modified
Remote Address	13.224.161.26:443
Referrer Policy	strict-origin-when-cross-origin

Response Headers	
Age	360
Date	Sat, 14 Feb 2026 16:12:56 GMT
Etag	"38be826f88f3f83b8fb118b9b32d64cc"
Server	AmazonS3
Via	1.1 e2b5386e09b1698ae0c3be879b94940a.cloudfront.net (CloudFront)
X-Amz-Cf-Id	bUvjBeUSZEiUw2OQoGt_ru82RZr7DXpX3px4g_xDOO42Ju1IAAYS0A==
X-Amz-Cf-Pop	DEL51-P5
X-Amz-Server-Side-Encryption	AES256
X-Cache	Hit from cloudfront

Request Headers	
:authority	d2v5y396zr42wz.cloudfront.net

5. Perform a cache invalidation and demonstrate content update immediately.

Ans:

I modified one of my static website files (for example, `index.html`) and re-uploaded it to the S3 bucket. Since CloudFront caches content, the old version was still being served.

To force CloudFront to fetch the updated content, I created a cache invalidation.

I specified the file path (or `/*` for all files). After the invalidation completed, I refreshed the website and verified that the updated content was served immediately.

The image shows two parts of a web development workflow. The top part is a screenshot of the AWS CloudFront console, specifically the 'Invalidation details' page for a distribution named 'E1P77O3NBUSMMU'. It shows an invalidation created on February 14, 2026, at 4:16:47 PM UTC, with a status of 'Completed'. The object paths are listed as '/*'. A 'Copy to new' button is visible. The bottom part is a screenshot of a web browser showing a 'Welcome to My Static Website' page, hosted on Amazon S3. The browser's developer tools are open to the 'Network' tab, showing a request to a CloudFront URL. The 'Headers' sub-tab is selected, displaying response headers from AmazonS3, including Date, Etag, Last-Modified, Server, Vary, Via, X-Amz-CF-Id, X-Amz-CF-Pop, X-Cache, and Request Headers.

Invalidation details

Date created
February 14, 2026 at 4:16:47 PM UTC

Status
✓ Completed

Object paths
/*

Welcome to My Static Website

Hosted on Amazon S3

Network

Filter: All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other

200 ms 400 ms 600 ms 800 ms 1,000 ms 1,200 ms

Headers

General

Request URL: https://d2v5y396zr42wz.cloudfront.net/
Request Method: GET
Status Code: 304 Not Modified
Remote Address: 13.224.161.26:443
Referrer Policy: strict-origin-when-cross-origin

Response Headers

Date: Sat, 14 Feb 2026 16:17:45 GMT
Etag: "38be826f88f3f83b8fb118b9b32d64cc"
Last-Modified: Sat, 14 Feb 2026 15:20:08 GMT
Server: AmazonS3
Vary: accept-encoding
Via: 1.1 cc35ef9ae729466b0fe3cddcf48ff56a.cloudfront.net (CloudFront)
X-Amz-CF-Id: 5erTkhkSoHz6oOO-MNaat5ewX70K4gVd57pT1e3DqnwaG4mSUuQ8w==
X-Amz-CF-Pop: DEL51-P5
X-Cache: Miss from cloudfront

Request Headers

6. Enable compression and validate reduced response size using browser developer tools.

Ans:

I enabled compression in my CloudFront distribution to reduce response size and improve performance. I configured the cache behavior to enable Gzip/Brotli compression for supported file types like HTML, CSS, and JavaScript. After deployment, I accessed my website and inspected response headers and file sizes in browser developer tools. I observed the Content-Encoding header showing gzip or br, and the transferred size was smaller compared to the original file size, confirming compression was working.

Edit behavior

Settings

Path pattern [Info](#)

/static/*

Origin and origin groups

plyush-bucket-2026-1.s3.eu-north-1.amazonaws.com-mlmgs8qh4y1

Compress objects automatically [Info](#)

- ☐ No
- ☒ Yes

Viewer

Viewer protocol policy

- ☐ HTTP and HTTPS
- ☒ Redirect HTTP to HTTPS
- ☐ HTTPS only

Allowed HTTP methods

- ☒ GET, HEAD
- ☐ GET, HEAD, OPTIONS
- ☐ GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE

Restrict viewer access

If you restrict viewer access, viewers must use CloudFront signed URLs or signed cookies to access your content.

- ☒ No
- ☐ Yes

Welcome to My Static Website

Hosted on Amazon S3

The screenshot shows the Network tab of a web browser's developer tools. A request to `d2v5y396zr42wz.cloudfront.net` has failed with a status of 403 (Forbidden). The request headers are visible, including `Accept`, `Accept-Encoding`, `Accept-Language`, `Cache-Control`, `If-Modified-Since`, `If-None-Match`, `Priority`, `Sec-Ch-Ua`, `Sec-Ch-Ua-Mobile`, `Sec-Ch-Ua-Platform`, `Sec-Fetch-Dest`, `Sec-Fetch-Mode`, `Sec-Fetch-Site`, `Sec-Fetch-User`, `Upgrade-Insecure-Requests`, and `User-Agent`.

Name	Value
:authority	d2v5y396zr42wz.cloudfront.net
:method	GET
:path	/
:scheme	https
Accept	text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding	gzip, deflate, br, zstd
Accept-Language	en-US,en;q=0.9
Cache-Control	max-age=0
If-Modified-Since	Sat, 14 Feb 2026 15:20:08 GMT
If-None-Match	"38be826f88f3f83b8fb118b9b32d64cc"
Priority	u=0, i
Sec-Ch-Ua	"Not(A:Brand";v="8", "Chromium";v="144", "Google Chrome";v="144"
Sec-Ch-Ua-Mobile	?0
Sec-Ch-Ua-Platform	"Linux"
Sec-Fetch-Dest	document
Sec-Fetch-Mode	navigate
Sec-Fetch-Site	none
Sec-Fetch-User	?1
Upgrade-Insecure-Requests	1
User-Agent	Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36

7. Attach a custom domain to CloudFront and configure SSL for HTTPS access.

Ans:

I configured a custom domain to access my CloudFront-hosted website. First, I requested an SSL certificate for my domain using AWS Certificate Manager (ACM) in the us-east-1 region. After validating domain ownership via DNS, I attached the certificate to my CloudFront distribution. I then added my custom domain as an Alternate Domain Name (CNAME) in CloudFront. Finally, I created a DNS record in Route 53 pointing my domain to the CloudFront distribution. I verified that my website was accessible securely over HTTPS using my custom domain.

The screenshot shows the AWS Route 53 console for the domain **siriusdevops.online**. The left sidebar lists navigation options: Route 53, Hosted zones, Health checks, Profiles, Global Resolver, VPC Resolver, Domains, IP-based routing, and Traffic flow. The main content area shows the 'Hosted zone details' for **siriusdevops.online**, including buttons for 'Delete zone', 'Test record', and 'Configure query logging'. Below this, the 'Records (4)' tab is active, displaying a table of DNS records.

Record ...	Type	Routin...	Differ...	Alias
☐ siriusdevo...	A	Simple	-	Yes
☐ siriusdevo...	NS	Simple	-	No
☐ siriusdevo...	SOA	Simple	-	No
☐ _e1c0ae4...	CNAME	Simple	-	No

Below the console, a browser window shows the website **Welcome to My Static Website** hosted on Amazon S3. The browser's address bar displays <https://siriusdevops.online>.

✔ Successfully updated distribution settings.

MyDistribution

Standard

View metrics

Details

Distribution domain name

d2v5y396zr42wz.cloudfront.net

Billing

Pay-as-you-go

Switch to a plan

ARN

arn:aws:cloudfront::596059883112:distribution/E1P77O3NBUSMMU

Last modified

Deploying

General

Security

Origins

Behaviors

Error pages

Invalidation

Logging

Tags

Settings

Edit

Name

MyDistribution

Description

-

Price class

Use all edge locations (best performance)

Supported HTTP versions

HTTP/2, HTTP/1.1, HTTP/1.0

Alternate domain names

[siriusdevops.online](#)

Route domains to CloudFront

Custom SSL certificate

[siriusdevops.online](#)

Security policy

TLSv1.3_2025

Standard logging

[Off](#)

Cookie logging

[Off](#)

Default root object

index.html

8. Apply geo-restriction to block access from two selected countries and verify behavior.

Ans:

I configured geo-restriction in my CloudFront distribution to block access from selected countries.

In the distribution settings, I enabled geographic restrictions and chose the Blacklist option. I added two countries to the blocked list. After deployment, I verified the configuration using a VPN from the restricted countries and received a 403 Forbidden response, confirming that access was successfully blocked.

Enable security protections to view the full security dashboard for this distribution

Gain visibility into top security trends, allowed and blocked traffic, as well as visibility and controls for bots. You'll later have the option to enable security logs to investigate traffic patterns—visually search, filter, and inspect individual log lines without writing queries.

[Learn more](#)

Manage security protections

▼ Security - Web Application Firewall (WAF)

Info

Manage protections

Keep your application secure from the most common web threats and security vulnerabilities using AWS WAF. Blocked requests are stopped before they reach your web servers.

Core protections

⊖ Not enabled

Advanced DDoS protection

⊖ Not enabled

Rate limiting

⊖ Not enabled

SQL protections

⊖ Not enabled

▼ CloudFront geographic restrictions

Restriction type

☐ No restrictions

☐ Allow list

☒ Block list

Countries

Select countries

China

Chile

Chad

Cancel

Save changes

piyush-buck x Distribution x Certificate x siriusdevop x ERROR: The x

d2v5y396zr42wz.cloudfront.net

403 ERROR

The request could not be satisfied.

The Amazon CloudFront distribution is configured to block access from your country. We can't connect to the server for this app or website at this time. There might be too much traffic or a configuration error. Try again later, or contact the app or website owner. If you provide content to customers through CloudFront, you can find steps to troubleshoot and help prevent this error by reviewing the CloudFront documentation.

Generated by cloudfront (CloudFront)
Request ID: S7VSdD1GcYPsgdS8TW0sU7qPSzVKq8J0CWnQcBHGmBqd6U5ABI9nmg==

9. Enable CloudFront access logs and identify cache hits, misses, client IPs, and status codes.

Ans:

I enabled access logging in my CloudFront distribution to monitor viewer requests. I configured an S3 bucket to store the logs and enabled standard logging in the distribution settings. After traffic was generated, log files were delivered to the bucket. I downloaded and analyzed the logs to identify cache hits, cache misses, client IP addresses, and HTTP status codes. This helped me understand request patterns and caching performance.

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Account and organization settings

AWS Marketplace for S3

cf-logs/

Copy S3 URI

Objects

Properties

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

1

	Name	Type	Last modified	Size	Storage class
	E1P77O3NB-USMMU.2026-02-14-16.e10657ce.gz	gz	February 14, 2026, 22:08:47 (UTC+05:30)	1.1 KB	Standard

```
[{"date": "2026-02-14", "time": "16:36:21", "x-edge-location": "DEL51-P5", "sc-bytes": "269", "c-ip": "103.68.31.203", "cs-method": "GET", "cs(host)": "d2v5v396zr42wz.cloudfront.net", "cs-uri-stem": "/", "sc-status": "304", "cs(referer)": "", "cs(user-agent)": "Mozilla/5.0%20(X11;%20Linux%20x86_64)%20AppleWebKit/537.36%20(KHTML,%20like%20Gecko)%20Chrome/144.0.0.0%20Safari/537.36", "cs-uri-query": "", "cs(cookie)": "", "x-edge-result-type": "Hit", "x-edge-request-id": "TtwqMM-ErtVrdkMydaAqIUWRI4i-uJmY-c9IezOA7-MoNwTOblaw=", "x-host-header": "d2v5v396zr42wz.cloudfront.net", "cs-protocol": "https", "cs-bytes": "29", "time-taken": "0.042", "x-forwarded-for": "", "ssl-protocol": "TLSv1.3", "ssl-cipher": "TLS_AES_128_GCM_SHA256", "x-edge-response-result-type": "Hit", "cs-protocol-version": "HTTP/2.0", "file-status": "", "file-encrypted-fields": "", "c-port": "12569", "time-to-first-byte": "0.042", "x-edge-detailed-result-type": "Hit", "sc-content-type": "", "sc-content-len": "", "sc-range-start": "", "sc-range-end": ""}, {"date": "2026-02-14", "time": "16:36:21", "x-edge-location": "DEL51-P5", "sc-bytes": "352", "c-ip": "103.68.31.203", "cs-method": "GET", "cs(host)": "d2v5v396zr42wz.cloudfront.net", "cs-uri-stem": "/favicon.ico", "sc-status": "403", "cs(referer)": "https://d2v5v396zr42wz.cloudfront.net/", "cs(user-agent)": "Mozilla/5.0%20(X11;%20Linux%20x86_64)%20AppleWebKit/537.36%20(KHTML,%20like%20Gecko)%20Chrome/144.0.0.0%20Safari/537.36", "cs-uri-query": "", "cs(cookie)": "", "x-edge-result-type": "Error", "x-edge-request-id": "qo8vZ9A06KsdzB6NoHuehZny3kUeIQWSSZB5tX3m-ukrIrhIA=", "x-host-header": "d2v5v396zr42wz.cloudfront.net", "cs-protocol": "https", "cs-bytes": "35", "time-taken": "0.185", "x-forwarded-for": "", "ssl-protocol": "TLSv1.3", "ssl-cipher": "TLS_AES_128_GCM_SHA256", "x-edge-response-result-type": "Error", "cs-protocol-version": "HTTP/2.0", "file-status": "", "file-encrypted-fields": "", "c-port": "12569", "time-to-first-byte": "0.185", "x-edge-detailed-result-type": "Error", "sc-content-type": "application/xml", "sc-content-len": "", "sc-range-start": "", "sc-range-end": ""}, {"date": "2026-02-14", "time": "16:36:19", "x-edge-location": "DEL51-P5", "sc-bytes": "353", "c-ip": "103.68.31.203", "cs-method": "GET", "cs(host)": "d2v5v396zr42wz.cloudfront.net", "cs-uri-stem": "/favicon.ico", "sc-status": "403", "cs(referer)": "https://d2v5v396zr42wz.cloudfront.net/", "cs(user-agent)": "Mozilla/5.0%20(X11;%20Linux%20x86_64)%20AppleWebKit/537.36%20(KHTML,%20like%20Gecko)%20Chrome/144.0.0.0%20Safari/537.36", "cs-uri-query": "", "cs(cookie)": "", "x-edge-result-type": "Error", "x-edge-request-id": "BVpQP6K3RBlcoQVDi2I6f-KI9Z4YduHDDYkVavChPxFnnszG-HTXQ=", "x-host-header": "d2v5v396zr42wz.cloudfront.net", "cs-protocol": "https", "cs-bytes": "35", "time-taken": "0.599", "x-forwarded-for": "", "ssl-protocol": "TLSv1.3", "ssl-cipher": "TLS_AES_128_GCM_SHA256", "x-edge-response-result-type": "Error", "cs-protocol-version": "HTTP/2.0", "file-status": "", "file-encrypted-fields": "", "c-port": "12569", "time-to-first-byte": "0.599", "x-edge-detailed-result-type": "Error", "sc-content-type": "application/xml", "sc-content-len": "", "sc-range-start": "", "sc-range-end": ""}, {"date": "2026-02-14", "time": "16:36:18", "x-edge-location": "DEL51-P5", "sc-bytes": "508", "c-ip": "103.68.31.203", "cs-method": "GET", "cs(host)": "d2v5v396zr42wz.cloudfront.net", "cs-uri-stem": "/", "sc-status": "200", "cs(referer)": "", "cs(user-agent)": "Mozilla/5.0%20(X11;%20Linux%20x86_64)%20AppleWebKit/537.36%20(KHTML,%20like%20Gecko)%20Chrome/144.0.0.0%20Safari/537.36", "cs-uri-query": "", "cs(cookie)": "", "x-edge-result-type": "Miss", "x-edge-request-id": "87C93e0i5BI1zH1I0TIO9uqV33TlqQVPRouBupNog4xHeXXU20=", "x-host-header": "d2v5v396zr42wz.cloudfront.net", "cs-protocol": "https", "cs-bytes": "27", "time-taken": "0.524", "x-forwarded-for": "", "ssl-protocol": "TLSv1.3", "ssl-cipher": "TLS_AES_128_GCM_SHA256", "x-edge-response-result-type": "Miss", "cs-protocol-version": "HTTP/2.0", "file-status": "", "file-encrypted-fields": "", "c-port": "12569", "time-to-first-byte": "0.524", "x-edge-detailed-result-type": "Miss", "sc-content-type": "text/html", "sc-content-len": "161", "sc-range-start": "", "sc-range-end": ""}, {"date": "2026-02-14", "time": "16:36:50", "x-edge-location": "DEL51-P5", "sc-bytes": "271", "c-ip": "103.68.31.203", "cs-method": "GET", "cs(host)": "d2v5v396zr42wz.cloudfront.net", "cs-uri-stem": "/", "sc-status": "304", "cs(referer)": "", "cs(user-agent)": "Mozilla/5.0%20(X11;%20Linux%20x86_64)%20AppleWebKit/537.36%20(KHTML,%20like%20Gecko)%20Chrome/144.0.0.0%20Safari/537.36", "cs-uri-query": "", "cs(cookie)": "", "x-edge-result-type": "Hit", "x-edge-request-id": "G-awWzScSHka0zB10TzdfOYANopLafvqZr3vBrI2IAoInkKlI2Iq=", "x-host-header": "d2v5v396zr42wz.cloudfront.net", "cs-protocol": "https", "cs-bytes": "29", "time-taken": "0.002", "x-forwarded-for": "", "ssl-protocol": "TLSv1.3", "ssl-cipher": "TLS_AES_128_GCM_SHA256", "x-edge-response-result-type": "Hit", "cs-protocol-version": "HTTP/2.0", "file-status": "", "file-encrypted-fields": "", "c-port": "12569", "time-to-first-byte": "0.002", "x-edge-detailed-result-type": "Hit", "sc-content-type": "", "sc-content-len": "", "sc-range-start": "", "sc-range-end": ""}, {"date": "2026-02-14", "time": "16:36:51", "x-edge-location": "DEL51-P5", "sc-bytes": "352", "c-ip": "103.68.31.203", "cs-method": "GET", "cs(host)": "d2v5v396zr42wz.cloudfront.net", "cs-uri-stem": "/favicon.ico", "sc-status": "403", "cs(referer)": "https://d2v5v396zr42wz.cloudfront.net/", "cs(user-agent)": "Mozilla/5.0%20(X11;%20Linux%20x86_64)%20AppleWebKit/537.36%20(KHTML,%20like%20Gecko)%20Chrome/144.0.0.0%20Safari/537.36", "cs-uri-query": "", "cs(cookie)": "", "x-edge-result-type": "Error", "x-edge-request-id": "crNo6m43Hv4UeIrD-5Ckxnu2Up608MRB17hCUBxydRXQ-ZUng=", "x-host-header": "d2v5v396zr42wz.cloudfront.net", "cs-protocol": "https", "cs-bytes": "35", "time-taken": "0.609", "x-forwarded-for": "", "ssl-protocol": "TLSv1.3", "ssl-cipher": "TLS_AES_128_GCM_SHA256", "x-edge-response-result-type": "Error", "cs-protocol-version": "HTTP/2.0", "file-status": "", "file-encrypted-fields": "", "c-port": "12569", "time-to-first-byte": "0.609", "x-edge-detailed-result-type": "Error", "sc-content-type": "application/xml", "sc-content-len": "", "sc-range-start": "", "sc-range-end": ""}]
```

10. Simulate a 403 error by modifying S3 permissions and then resolve it.

Ans:

I simulated a 403 Access Denied error by modifying the S3 bucket permissions. I removed the CloudFront Origin Access Control permission from the bucket policy, which prevented CloudFront from accessing objects. When I accessed the website through CloudFront, I received a 403 Forbidden error. To resolve the issue, I restored the correct bucket policy allowing access only from the CloudFront distribution. After redeployment, I verified that the website was accessible again.

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
</Error>
```



11. Create a scenario where content is not cached and fix the caching configuration.

Ans:

I created a scenario where content was not being cached by setting the cache TTL values to zero in the CloudFront cache behavior. Because caching was disabled, every request was forwarded to the origin, and response headers showed Miss from CloudFront each time. To fix this, I modified the cache policy and configured appropriate TTL values. After redeployment, I verified caching again using response headers, which showed Hit from CloudFront, confirming that caching was working correctly.

CloudFront > Policies > Cache > 93eba422-fb44-46a8-b826-d7a29842fab1

static-policy

Edit Delete

Details

Description	ID	Last modified
-	93eba422-fb44-46a8-b826-d7a29842fab1	February 14, 2026 at 4:49:49 PM UTC

TTL settings

Minimum TTL (seconds)	Maximum TTL (seconds)	Default TTL (seconds)
1	1	1

Cache key settings

Headers - None	Cookies - None	Query strings - None
----------------	----------------	----------------------

Welcome to My Static Website

Hosted on Amazon S3

The screenshot shows the Network tab of a web browser. A request for `favicon.ico` is selected. The response headers are as follows:

Header	Value
Date	Sat, 14 Feb 2026 16:50:51 GMT
Etag	"38be826f88f3f83b8fb118b9b32d64cc"
Last-Modified	Sat, 14 Feb 2026 15:20:08 GMT
Server	AmazonS3
Vary	accept-encoding
Via	1.1 93fcc80e2bee004e8dc8b0fa4432ff7c.cloudfront.net (CloudFront)
X-Amz-Cf-Id	nqBwVHiPZTXsjR1e88YVlcHVMVeG8XzwaZagf6Jpr01nOqJ5couHQw==
X-Amz-Cf-Pop	DEL51-P5
X-Cache	Miss from cloudfront

12. Configure two origins (S3 and ALB) and route `/api/*` traffic to ALB using path-based routing.

Ans:

I configured my CloudFront distribution with two origins: an S3 bucket for static website content and an Application Load Balancer (ALB) for API traffic. I added the ALB DNS name

as a second origin in CloudFront. Then, I created a cache behavior with the path pattern `/api/*` and mapped it to the ALB origin. This enabled path-based routing where static content was served from S3 and API requests were forwarded to the ALB backend. I verified the configuration by accessing `/api/*` endpoints and confirming responses were served from the ALB.



Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager [New](#)

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups

testing



Actions ▼

▼ Details

Load balancer type

Application

Scheme

Internet-facing

VPC

[vpc-0b10c092f8079ece5](#)

Availability Zones

[subnet-035855527f66b6233](#) eu-north-1a (eun1-az1)

[subnet-04d4cf52ea5c56dbc](#) eu-north-1b (eun1-az2)

Load balancer ARN

`arn:aws:elasticloadbalancing:eu-north-1:596059883112:loadbalancer/app/testing/60b6d6aa4223e9ad`

Status

Provisioning

Hosted zone

Z23TAZ6LKFMNIO

Load balancer IP address type

IPv4

Date created

February 14, 2026, 22:24 (UTC+05:30)

DNS name copied

`testing-1350290890.eu-north-1.elb.amazonaws.com` (A Record)



Listeners and rules

Network mapping

Resource map

Security groups



Listeners and rules (1) [Info](#)



Manage rules ▼

Manage listener ▼

Add listener



✓ Successfully created new cache behavior /api/*.

MyDistribution Standard

[View metrics](#)

Details

Distribution domain name
 d2v5y396zr42wz.cloudfront.net

Billing
Pay-as-you-go
[Switch to a plan](#)

ARN
 arn:aws:cloudfront::596059883112:distribution/E1P77O3NBUSMMU

Last modified
 Deploying

- General
- Security
- Origins
- Behaviors**
- Error pages
- Invalidations
- Logging
- Tags

Behaviors (3) Save Move up Move down Edit Delete Create behavior

	Preced...	Path patt...	Origin or ...	Viewer pr...	Cache pol...	Origin req...	Response ...
☐	0	/static/*	piyush-bu...	Redirect H...	static-policy	-	-
☐	1	/api/*	testing-13...	Redirect H...	Managed-Cach...	Managed-AllVie	-
☐	2	Default (*)	piyush-bu...	Redirect H...	Managed-Cach...	-	-

